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Reducing throughput of paired cameras

Abstract

Reducing throughput of paired cameras, including: selecting, by a first camera of an autonomous vehicle, a first area of focus for a first frame generated by the first camera; selecting, by a second camera of the autonomous vehicle, a second area of focus for a second frame generated by the second camera by matching the first area of focus for the first frame; generating, by the first camera from the first frame, a first downsampled frame and a first cropped frame, wherein the first cropped frame is based on the first area of focus; and generating, by the second camera from the second frame, a second downsampled frame and a second cropped frame, wherein the second cropped frame is based on the second area of focus.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This is a continuation application for patent entitled to a filing date and claiming the benefit of earlier-filed U.S. patent application Ser. No. 17/676,569, filed Feb. 21, 2022, herein incorporated by reference in its entirety.

BACKGROUND

Field of the Invention

(1) The field of the invention is autonomous vehicle systems, or, more specifically, methods, apparatus, autonomous vehicles, and products for throughput reduction in high resolution camera sensors.

SUMMARY

(2) Throughput reduction in autonomous vehicle camera sensors may include: generating, by a camera sensor, a frame; selecting an area of focus for the frame; and generating, by the camera sensor from the frame, a downsampled frame and a cropped frame, wherein the cropped frame is based on the area of focus.

(3) The foregoing and other objects, features and advantages of the invention will be apparent from the following more particular descriptions of exemplary embodiments of the invention as illustrated in the accompanying drawings wherein like reference numbers generally represent like parts of exemplary embodiments of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows example views of an autonomous vehicle for throughput reduction in high resolution camera sensors according to some embodiments of the present disclosure.

(2) FIG. 2 is a block diagram of an autonomous computing system for throughput reduction in high resolution camera sensors according to some embodiments of the present disclosure.

(3) FIG. 3 is a block diagram of a redundant power fabric for throughput reduction in high resolution camera sensors according to some embodiments of the present disclosure.

(4) FIG. 4 is a block diagram of a redundant data fabric for throughput reduction in high resolution camera sensors according to some embodiments of the present disclosure.

(5) FIG. 5 is an example view of process allocation across CPU packages for throughput reduction in high resolution camera sensors according to some embodiments of the present disclosure.

(6) FIG. 6 is an example view of an execution environment for throughput reduction in high resolution camera sensors according to some embodiments of the present disclosure.

(7) FIG. 7 is a flowchart of an example method for throughput reduction in high resolution camera sensors according to some embodiments of the present disclosure.

(8) FIG. 8 is a flowchart of an example method for throughput reduction in high resolution camera sensors according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

(9) The terminology used herein for the purpose of describing particular examples is not intended to be limiting for further examples. Whenever a singular form such as “a”, “an” and “the” is used and using only a single element is neither explicitly or implicitly defined as being mandatory, further examples may also use plural elements to implement the same functionality. Likewise, when a functionality is subsequently described as being implemented using multiple elements,

further examples may implement the same functionality using a single element or processing entity. It will be further understood that the terms “comprises”, “comprising”, “includes” and/or “including”, when used, specify the presence of the stated features, integers, steps, operations, processes, acts, elements and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, processes, acts, elements, components and/or any group thereof. Additionally, when an element is described as “plurality,” it is understood to mean two or more of such an element. However, as set forth above, further examples may implement the same functionality using a single element.

(10) It will be understood that when an element is referred to as being “connected” or “coupled” to another element, the elements may be directly connected or coupled or via one or more intervening elements. If two elements A and B are combined using an “or”, this is to be understood to disclose all possible combinations, i.e. only A, only B, as well as A and B. An alternative wording for the same combinations is “at least one of A and B”. The same applies for combinations of more than two elements.

(11) Accordingly, while further examples are capable of various modifications and alternative forms, some particular examples thereof are shown in the figures and will subsequently be described in detail. However, this detailed description does not limit further examples to the particular forms described. Further examples may cover all modifications, equivalents, and alternatives falling within the scope of the disclosure. Like numbers refer to like or similar elements throughout the description of the figures, which may be implemented identically or in modified form when compared to one another while providing for the same or a similar functionality.

(12) Throughput reduction in autonomous vehicle camera sensors may be implemented in an autonomous vehicle. Accordingly, FIG. 1 shows multiple views of an autonomous vehicle **100** configured for throughput reduction in high resolution camera sensors according to embodiments of the present invention. Right side view **101a** shows a right side of the autonomous vehicle **100**. Shown in the right side view **101a** are cameras **102** and **103**, configured to capture image data, video data, and/or audio data of the environmental state of the autonomous vehicle **100** from the perspective of the right side of the car. Front view **101b** shows a front side of the autonomous vehicle **100**. Shown in the front view **101b** are cameras **104** and **106**, configured to capture image data, video data, and/or audio data of the environmental state of the autonomous vehicle **100** from the perspective of the front of the car. Rear view **101c** shows a rear side of the autonomous vehicle **100**. Shown in the rear view **101c** are cameras **108** and **110**, configured to capture image data, video data, and/or audio data of the environmental state of the autonomous vehicle **100** from the perspective of the rear of the car. Top view **101d** shows a rear side of the autonomous vehicle **100**. Shown in the top view **101d** are cameras **102-110**. Also shown are cameras **112** and **114**, configured to capture image data, video data, and/or audio data of the environmental state of the autonomous vehicle **100** from the perspective of the left side of the car.

(13) Further shown in the top view **101d** is an automation computing system **116**. The automation computing system **116** comprises one or more computing devices configured to control one or more autonomous operations (e.g., autonomous driving operations) of the autonomous vehicle **100**. For example, the automation computing system **116** may be configured to process sensor data (e.g., data from the cameras **102-114** and potentially other sensors), operational data (e.g., a speed, acceleration, gear, orientation, turning direction), and other data to determine a operational state and/or operational history of the autonomous vehicle. The automation computing system **116** may then determine one or more operational commands for the autonomous vehicle (e.g., a change in speed or acceleration, a change in brake application, a change in gear, a change in turning or orientation, etc.). The automation computing system **116** may also capture and store sensor data. Operational data of the autonomous vehicle may also be stored in association with corresponding sensor data, thereby indicating the operational data of the autonomous vehicle **100** at the time the

sensor data was captured.

(14) Although the autonomous vehicle **100** of FIG. **1** is shown as car, it is understood that autonomous vehicles **100** configured for throughput reduction in high resolution camera sensors may also include other vehicles, including motorcycles, planes, helicopters, unmanned aerial vehicles (UAVs, e.g., drones), or other vehicles as can be appreciated. Moreover, it is understood that additional cameras or other external sensors may also be included in the autonomous vehicle **100**.

(15) Throughput reduction in autonomous vehicle camera sensors in accordance with the present invention is generally implemented with computers, that is, with automated computing machinery. For further explanation, therefore, FIG. **2** sets forth a block diagram of automated computing machinery comprising an exemplary automation computing system **116** configured for throughput reduction in high resolution camera sensors according to embodiments of the present invention. The automation computing system **116** of FIG. **2** includes at least one computer Central Processing Unit (CPU) package **204** as well as random access memory **206** ('RAM') which is connected through a high speed memory bus **208** and bus adapter **210** to CPU packages **204** via a front side bus **211** and to other components of the automation computing system **116**.

(16) A CPU package **204** may comprise a plurality of processing units. For example, each CPU package **204** may comprise a logical or physical grouping of a plurality of processing units. Each processing unit may be allocated a particular process for execution. Moreover, each CPU package **204** may comprise one or more redundant processing units. A redundant processing unit is a processing unit not allocated a particular process for execution unless a failure occurs in another processing unit. For example, when a given processing unit allocated a particular process fails, a redundant processing unit may be selected and allocated the given process. A process may be allocated to a plurality of processing units within the same CPU package **204** or different CPU packages **204**. For example, a given process may be allocated to a primary processing unit in a CPU package **204**. The results or output of the given process may be output from the primary processing unit to a receiving process or service. The given process may also be executed in parallel on a secondary processing unit. The secondary processing unit may be included within the same CPU package **204** or a different CPU package **204**. The secondary processing unit may not provide its output or results of the process until the primary processing unit fails. The receiving process or service will then receive data from the secondary processing unit. A redundant processing unit may then be selected and have allocated the given process to ensure that two or more processing units are allocated the given process for redundancy and increased reliability.

(17) The CPU packages **204** are communicatively coupled to one or more sensors **212**. The sensors **212** are configured to capture sensor data describing the operational and environmental conditions of an autonomous vehicle. For example, the sensors **212** may include cameras (e.g., the cameras **102-114** of FIG. **1**), accelerometers, Global Positioning System (GPS) radios, Lidar sensors, or other sensors as can be appreciated. As described herein, cameras may include a solid state sensor **212** with a solid state shutter capable of measuring photons or a time of flight of photons. For example, a camera may be configured to capture or measure photons captured via the shutter for encoding as images and/or video data. As another example, a camera may emit photons and measure the time of flight of the emitted photons. Cameras may also include event cameras configured to measure changes in light and/or motion of light.

(18) Although the sensors **212** are shown as being external to the automation computing system **116**, it is understood that one or more of the sensors **212** may reside as a component of the automation computing system **116** (e.g., on the same board, within the same housing or chassis). The sensors **212** may be communicatively coupled with the CPU packages **204** via a switched fabric **213**. The switched fabric **213** comprises a communications topology through which the CPU packages **204** and sensors **212** are coupled via a plurality of switching mechanisms (e.g., latches, switches, crossbar switches, field programmable gate arrays (FPGAs), etc.). For example, the

switched fabric **213** may implement a mesh connection connecting the CPU packages **204** and sensors **212** as endpoints, with the switching mechanisms serving as intermediary nodes of the mesh connection. The CPU packages **204** and sensors **212** may be in communication via a plurality of switched fabrics **213**. For example, each of the switched fabrics **213** may include the CPU packages **204** and sensors **212**, or a subset of the CPU packages **204** and sensors **212**, as endpoints. Each switched fabric **213** may also comprise a respective plurality of switching components. The switching components of a given switched fabric **213** may be independent (e.g., not connected) of the switching components of other switched fabrics **213** such that only switched fabric **213** endpoints (e.g., the CPU packages **204** and sensors **212**) are overlapping across the switched fabrics **213**. This provides redundancy such that, should a connection between a CPU package **204** and sensor **212** fail in one switched fabric **213**, the CPU package **204** and sensor **212** may remain connected via another switched fabric **213**. Moreover, in the event of a failure in a CPU package **204**, a processor of a CPU package **204**, or a sensor, a communications path excluding the failed component and including a functional redundant component may be established.

(19) The CPU packages **204** and sensors **212** are configured to receive power from one or more power supplies **215**. The power supplies **215** may comprise an extension of a power system of the autonomous vehicle **100** or an independent power source (e.g., a battery). The power supplies **215** may supply power to the CPU packages **204** and sensors **212** by another switched fabric **214**. The switched fabric **214** provides redundant power pathways such that, in the event of a failure in a power connection, a new power connection pathway may be established to the CPU packages **204** and sensors **212**.

(20) Stored in RAM **206** is an automation module **220**. The automation module **220** may be configured to process sensor data from the sensors **212** to determine a driving decision for the autonomous vehicle. The driving decision comprises one or more operational commands for an autonomous vehicle **100** to affect the movement, direction, or other function of the autonomous vehicle **100**, thereby facilitating autonomous driving or operation of the vehicle. Such operational commands may include a change in the speed of the autonomous vehicle **100**, a change in steering direction, a change in gear, or other command as can be appreciated. For example, the automation module **220** may provide sensor data and/or processed sensor data as one or more inputs to a trained machine learning model (e.g., a trained neural network) to determine the one or more operational commands. The operational commands may then be communicated to autonomous vehicle control systems **223** via a vehicle interface **222**.

(21) In some embodiments, the automation module **220** may be configured to determine an exit path for an autonomous vehicle **100** in motion. The exit path includes one or more operational commands that, if executed, are determined and/or predicted to bring the autonomous vehicle **100** safely to a stop (e.g., without collision with an object, without violating one or more safety rules). The automation module **220** may determine both a driving decision and an exit path at a predefined interval. The automation module **220** may then send the driving decision and the exit path to the autonomous vehicle control systems **223**. The autonomous vehicle control systems **223** may be configured to execute the driving decision unless an error state has been reached. If an error decision has been reached, therefore indicating a possible error in functionality of the automation computing system **116**), the autonomous vehicle control systems **223** may then execute a last received exit path in order to bring the autonomous vehicle **100** safely to a stop. Thus, the autonomous vehicle control systems **223** are configured to receive both a driving decision and exit path at predefined intervals, and execute the exit path in response to an error.

(22) The autonomous vehicle control systems **223** are configured to affect the movement and operation of the autonomous vehicle **100**. For example, the autonomous vehicle control systems **223** may activate (e.g., apply one or more control signals) to actuators or other components to turn or otherwise change the direction of the autonomous vehicle **100**, accelerate or decelerate the autonomous vehicle **100**, change a gear of the autonomous vehicle **100**, or otherwise affect the

movement and operation of the autonomous vehicle **100**.

(23) Further stored in RAM **206** is a data collection module **224** configured to process and/or store sensor data received from the one or more sensors **212**. For example, the data collection module **224** may store the sensor data as captured by the one or more sensors **212**, or processed sensor **212** data (e.g., sensor **212** data having object recognition, compression, depth filtering, or other processes applied). Such processing may be performed by the data collection module **224** in real-time or in substantially real-time as the sensor data is captured by the one or more sensors **212**. The processed sensor data may then be used by other functions or modules. For example, the automation module **220** may use processed sensor data as input to determine one or more operational commands. The data collection module **224** may store the sensor data in data storage **218**.

(24) Also stored in RAM **206** is a data processing module **226**. The data processing module **226** is configured to perform one or more processes on stored sensor data (e.g., stored in data storage **218** by the data collection module **218**) prior to upload to an execution environment **227**. Such operations can include filtering, compression, encoding, decoding, or other operations as can be appreciated. The data processing module **226** may then communicate the processed and stored sensor data to the execution environment **227**.

(25) Further stored in RAM **206** is a hypervisor **228**. The hypervisor **228** is configured to manage the configuration and execution of one or more virtual machines **229**. For example, each virtual machine **229** may emulate and/or simulate the operation of a computer. Accordingly, each virtual machine **229** may comprise a guest operating system **216** for the simulated computer. The hypervisor **228** may manage the creation of a virtual machine **229** including installation of the guest operating system **216**. The hypervisor **228** may also manage when execution of a virtual machine **229** begins, is suspended, is resumed, or is terminated. The hypervisor **228** may also control access to computational resources (e.g., processing resources, memory resources, device resources) by each of the virtual machines.

(26) Each of the virtual machines **229** may be configured to execute one or more of the automation module **220**, the data collection module **224**, the data processing module **226**, or combinations thereof. Moreover, as is set forth above, each of the virtual machines **229** may comprise its own guest operating system **216**. Guest operating systems **216** useful in autonomous vehicles in accordance with some embodiments of the present disclosure include UNIX™, Linux™, Microsoft Windows™, AIX™, IBM's i OS™, and others as will occur to those of skill in the art. For example, the autonomous vehicle **100** may be configured to execute a first operating system when the autonomous vehicle is in an autonomous (or even partially autonomous) driving mode and the autonomous vehicle **100** may be configured to execute a second operating system when the autonomous vehicle is not in an autonomous (or even partially autonomous) driving mode. In such an example, the first operating system may be formally verified, secure, and operate in real-time such that data collected from the sensors **212** are processed within a predetermined period of time, and autonomous driving operations are performed within a predetermined period of time, such that data is processed and acted upon essentially in real-time. Continuing with this example, the second operating system may not be formally verified, may be less secure, and may not operate in real-time as the tasks that are carried out (which are described in greater detail below) by the second operating system are not as time-sensitive the tasks (e.g., carrying out self-driving operations) performed by the first operating system.

(27) Readers will appreciate that although the example included in the preceding paragraph relates to an embodiment where the autonomous vehicle **100** may be configured to execute a first operating system when the autonomous vehicle is in an autonomous (or even partially autonomous) driving mode and the autonomous vehicle **100** may be configured to execute a second operating system when the autonomous vehicle is not in an autonomous (or even partially autonomous) driving mode, other embodiments are within the scope of the present disclosure. For example, in

another embodiment one CPU (or other appropriate entity such as a chip, CPU core, and so on) may be executing the first operating system and a second CPU (or other appropriate entity) may be executing the second operating system, where switching between these two modalities is accomplished through fabric switching, as described in greater detail below. Likewise, in some embodiments, processing resources such as a CPU may be partitioned where a first partition supports the execution of the first operating system and a second partition supports the execution of the second operating system.

(28) The guest operating systems **216** may correspond to a particular operating system modality. An operating system modality is a set of parameters or constraints which a given operating system satisfies, and are not satisfied by operating systems of another modality. For example, a given operating system may be considered a “real-time operating system” in that one or more processes executed by the operating system must be performed according to one or more time constraints. For example, as the automation module **220** must make determinations as to operational commands to facilitate autonomous operation of a vehicle. Accordingly, the automation module **220** must make such determinations within one or more time constraints in order for autonomous operation to be performed in real time. The automation module **220** may then be executed in an operating system (e.g., a guest operating system **216** of a virtual machine **229**) corresponding to a “real-time operating system” modality. Conversely, the data processing module **226** may be able to perform its processing of sensor data independent of any time constraints, and may then be executed in an operating system (e.g., a guest operating system **216** of a virtual machine **229**) corresponding to a “non-real-time operating system” modality.

(29) As another example, an operating system (e.g., a guest operating system **216** of a virtual machine **229**) may comprise a formally verified operating system. A formally verified operating system is an operating system for which the correctness of each function and operation has been verified with respect to a formal specification according to formal proofs. A formally verified operating system and an unverified operating system (e.g., one that has not been formally verified according to these proofs) can be said to operate in different modalities.

(30) The automation module **220**, data collection module **224**, data collection module **224**, data processing module **226**, hypervisor **228**, and virtual machine **229** in the example of FIG. 2 are shown in RAM **206**, but many components of such software typically are stored in non-volatile memory also, such as, for example, on data storage **218**, such as a disk drive. Moreover, any of the automation module **220**, data collection module **224**, and data processing module **226** may be executed in a virtual machine **229** and facilitated by a guest operating system **216** of that virtual machine **229**.

(31) The automation computing system **116** of FIG. 2 includes disk drive adapter **230** coupled through expansion bus **232** and bus adapter **210** to CPU package(s) **204** and other components of the automation computing system **116**. Disk drive adapter **230** connects non-volatile data storage to the automation computing system **116** in the form of data storage **218**. Disk drive adapters **230** useful in computers configured for throughput reduction in high resolution camera sensors according to embodiments of the present invention include Integrated Drive Electronics (‘IDE’) adapters, Small Computer System Interface (‘SCSI’) adapters, and others as will occur to those of skill in the art. Non-volatile computer memory also may be implemented for as an optical disk drive, electrically erasable programmable read-only memory (so-called ‘EEPROM’ or ‘Flash’ memory), RAM drives, and so on, as will occur to those of skill in the art.

(32) The exemplary automation computing system **116** of FIG. 2 includes a communications adapter **238** for data communications with other computers and for data communications with a data communications network. Such data communications may be carried out serially through RS-232 connections, through external buses such as a Universal Serial Bus (‘USB’), through data communications networks such as IP data communications networks, and in other ways as will occur to those of skill in the art. Communications adapters implement the hardware level of data

communications through which one computer sends data communications to another computer, directly or through a data communications network. Examples of communications adapters useful in computers configured for throughput reduction in high resolution camera sensors according to embodiments of the present invention include modems for wired dial-up communications, Ethernet (IEEE 802.3) adapters for wired data communications, 802.11 adapters for wireless data communications, as well as mobile adapters (e.g., cellular communications adapters) for mobile data communications. For example, the automation computing system **116** may communicate with one or more remotely disposed execution environments **227** via the communications adapter **238**. (33) The exemplary automation computing system of FIG. **2** also includes one or more Artificial Intelligence (AI) accelerators **240**. The AI accelerator **240** provides hardware-based assistance and acceleration of AI-related functions, including machine learning, computer vision, etc. Accordingly, performance of any of the automation module **220**, data collection module **224**, data processing module **226**, or other operations of the automation computing system **116** may be performed at least in part by the AI accelerators **240**.

(34) The exemplary automation computing system of FIG. **2** also includes one or more graphics processing units (GPUs) **242**. The GPUs **242** are configured to provide additional processing and memory resources for processing image and/or video data, including encoding, decoding, etc. Accordingly, performance of any of the automation module **220**, data collection module **224**, data processing module **226**, or other operations of the automation computing system **116** may be performed at least in part by the GPUs **242**.

(35) FIG. **3** shows an example redundant power fabric for throughput reduction in high resolution camera sensors. The redundant power fabric provides redundant pathways for power transfer between the power supplies **215**, the sensors **212**, and the CPU packages **204**. In this example, the power supplies **215** are coupled to the sensors **212** and CPU packages via two switched fabrics **214a** and **214b**. The topology shown in FIG. **3** provides redundant pathways between the power supplies **215**, the sensors **212**, and the CPU packages **204** such that power can be rerouted through any of multiple pathways in the event of a failure in an active connection pathway. The switched fabrics **214a** and **214b** may provide power to the sensors **212** using various connections, including Mobile Industry Processor Interface (MIPI), Inter-Integrated Circuit (I2C), Universal Serial Bus (USB), or another connection. The switched fabrics **214a** and **214b** may also provide power to the CPU packages **204** using various connections, including Peripheral Component Interconnect Express (PCIe), USB, or other connections. Although only two switched fabrics **214a** and **214b** are shown connecting the power supplies **215** to the sensors **212** and CPU packages **204**, it is understood that the approach shown by FIG. **3** can be modified to include additional switched fabrics **214**.

(36) FIG. **4** is an example redundant data fabric for throughput reduction in high resolution camera sensors. The redundant data fabric provides redundant data connection pathways between sensors **212** and CPU packages **204**. In this example view, three CPU packages **204a**, **204b**, and **204c** are connected to three sensors **212a**, **212b**, and **212c** via three switched fabrics **213a**, **213b**, and **213c**. Each CPU package **204a**, **204b**, and **204c** is connected to a subset of the switched fabrics **213a**, **213b**, and **213c**. For example, CPU package **204a** is connected to switched fabrics **213a** and **213c**, CPU package **204b** is connected to switched fabrics **213a** and **213b**, and CPU package **204c** is connected to switched fabrics **213b** and **213c**. Each switched fabric **213a**, **213b**, and **213c** is connected to a subset of the sensors **212a**, **212b**, and **212c**. For example, switched fabric **213a** is connected to sensors **212a** and **212b**, switched fabric **213b** is connected to sensor **212b** and **212c**, and switched fabric **213c** is connected to sensors **212a** and **212c**. Under this topology, each CPU package **204a**, **204b**, and **204c** has an available connection path to any sensor **212a**, **212b**, and **212c**. It is understood that the topology of FIG. **4** is exemplary, and that CPU packages, switched fabrics, sensors, or connections between components may be added or removed while maintaining redundancy as can be appreciated by one skilled in the art.

(37) FIG. 5 is an example view of process allocation across CPU packages for throughput reduction in high resolution camera sensors. Shown are three CPU packages **204a**, **204b**, and **204c**. Each CPU package **204a** includes a processing unit that has been allocated (e.g., by a hypervisor **228** or other process or service) primary execution of a process and another processing unit that has been allocated secondary execution of a process. As set forth herein, primary execution of a process describes an executing instance of a process whose output will be provided to another process or service. Secondary execution of the process describes executing an instance of the process in parallel to the primary execution, but the output may not be output to the other process or service. For example, in CPU package **204a**, processing unit **502a** has been allocated secondary execution of “process B,” denoted as secondary process B **504b**, while processing unit **502b** has been allocated primary execution of “process C,” denoted as primary process C **506a**.

(38) CPU package **204a** also comprises two redundant processing units that are not actively executing a process A, B, or C, but are instead reserved in case of failure of an active processing unit. Redundant processing unit **508a** has been reserved as “A/B redundant,” indicating that reserved processing unit **508a** may be allocated primary or secondary execution of processes A or B in the event of a failure of a processing unit allocated the primary or secondary execution of these processes. Redundant processing unit **508b** has been reserved as “A/C redundant,” indicating that reserved processing unit **508b** may be allocated primary or secondary execution of processes A or C in the event of a failure of a processing unit allocated the primary or secondary execution of these processes.

(39) CPU package **204b** includes processing unit **502c**, which has been allocated primary execution of “process A,” denoted as primary process A **510a**, and processing unit **502d**, which has been allocated secondary execution of “process C,” denoted as secondary process C **506a**. CPU package **204b** also includes redundant processing unit **508c**, reserved as “A/B redundant,” and redundant processing unit **508d**, reserved as “B/C redundant.” CPU package **204c** includes processing unit **502e**, which has been allocated primary execution of “process B,” denoted as primary process B **504a**, and processing unit **502f**, which has been allocated secondary execution of “process A,” denoted as secondary process A **510b**. CPU package **204c** also includes redundant processing unit **508e**, reserved as “B/C redundant,” and redundant processing unit **508f**, reserved as “A/C redundant.”

(40) As set forth in the example view of FIG. 5, primary and secondary instances processes A, B, and C are each executed in an allocated processing unit. Thus, if a processing unit performing primary execution of a given process fails, the processing unit performing secondary execution may instead provide output of the given process to a receiving process or service. Moreover, the primary and secondary execution of a given process are executed on different CPU packages. Thus, if an entire processing unit fails, execution of each of the processes can continue using one or more processing units handling secondary execution. The redundant processing units **508a-f** allow for allocation of primary or secondary execution of a process in the event of processing unit failure. This further prevents errors caused by processing unit failure as parallel primary and secondary execution of a process may be restored. One skilled in the art would understand that the number of CPU packages, processing units, redundant processing units, and processes may be modified according to performance requirements while maintaining redundancy.

(41) For further explanation, FIG. 6 sets forth a diagram of an execution environment **227** accordance with some embodiments of the present disclosure. The execution environment **227** depicted in FIG. 6 may be embodied in a variety of different ways. The execution environment **227** may be provided, for example, by one or more physical or virtual machine components consisting of bare-metal applications, operating systems such as Android, Linux, Real-time Operating systems (RTOS), Automotive RTOS, such as AutoSAR, and others, including combinations thereof. The execution environment **227** may also be provided by cloud computing providers such as Amazon AWS, Microsoft Azure, Google Cloud, and others, including combinations thereof. Alternatively,

the execution environment **227** may be embodied as a collection of devices (e.g., servers, storage devices, networking devices) and software resources that are included in a computer or distributed computer or private data center. Readers will appreciate that the execution environment **227** may be constructed in a variety of other ways and may even include resources within one or more autonomous vehicles or resources that communicate with one or more autonomous vehicles.

(42) The execution environment **227** depicted in FIG. **6** may include storage resources **608**, which may be embodied in many forms. For example, the storage resources **608** may include flash memory, hard disk drives, nano-RAM, 3D crosspoint non-volatile memory, MRAM, non-volatile phase-change memory ('PCM'), storage class memory ('SCM'), or many others, including combinations of the storage technologies described above. Readers will appreciate that other forms of computer memories and storage devices may be utilized as part of the execution environment **227**, including DRAM, SRAM, EEPROM, universal memory, and many others. The storage resources **608** may also be embodied, in embodiments where the execution environment **227** includes resources offered by a cloud provider, as cloud storage resources such as Amazon Elastic Block Storage ('EBS') block storage, Amazon S3 object storage, Amazon Elastic File System ('EFS') file storage, Azure Blob Storage, and many others. The example execution environment **227** depicted in FIG. **6** may implement a variety of storage architectures, such as block storage where data is stored in blocks, and each block essentially acts as an individual hard drive, object storage where data is managed as objects, or file storage in which data is stored in a hierarchical structure. Such data may be saved in files and folders, and presented to both the system storing it and the system retrieving it in the same format.

(43) The execution environment **227** depicted in FIG. **6** also includes communications resources **610** that may be useful in facilitating data communications between components within the execution environment **227**, as well as data communications between the execution environment **227** and computing devices that are outside of the execution environment **227**. Such communications resources may be embodied, for example, as one or more routers, network switches, communications adapters, and many others, including combinations of such devices. The communications resources **610** may be configured to utilize a variety of different protocols and data communication fabrics to facilitate data communications. For example, the communications resources **610** may utilize Internet Protocol ('IP') based technologies, fibre channel ('FC') technologies, FC over ethernet ('FCOE') technologies, InfiniBand ('IB') technologies, NVMe Express ('NVMe') technologies and NVMe over fabrics ('NVMeoF') technologies, and many others. The communications resources **610** may also be embodied, in embodiments where the execution environment **227** includes resources offered by a cloud provider, as networking tools and resources that enable secure connections to the cloud as well as tools and resources (e.g., network interfaces, routing tables, gateways) to configure networking resources in a virtual private cloud. Such communications resources may be useful in facilitating data communications between components within the execution environment **227**, as well as data communications between the execution environment **227** and computing devices that are outside of the execution environment **227** (e.g., computing devices that are included within an autonomous vehicle).

(44) The execution environment **227** depicted in FIG. **6** also includes processing resources **612** that may be useful in useful in executing computer program instructions and performing other computational tasks within the execution environment **227**. The processing resources **612** may include one or more application-specific integrated circuits ('ASICs') that are customized for some particular purpose, one or more central processing units ('CPUs'), one or more digital signal processors ('DSPs'), one or more field-programmable gate arrays ('FPGAs'), one or more systems on a chip ('SoCs'), or other form of processing resources **612**. The processing resources **612** may also be embodied, in embodiments where the execution environment **227** includes resources offered by a cloud provider, as cloud computing resources such as one or more Amazon Elastic Compute Cloud ('EC2') instances, event-driven compute resources such as AWS Lambdas, Azure

Virtual Machines, or many others.

(45) The execution environment **227** depicted in FIG. **6** also includes software resources **613** that, when executed by processing resources **612** within the execution environment **227**, may perform various tasks. The software resources **613** may include, for example, one or more modules of computer program instructions that when executed by processing resources **612** within the execution environment **227** are useful in training neural networks configured to determine control autonomous vehicle control operations. For example, a training module **614** may train a neural network using training data including sensor **212** data and control operations recorded or captured contemporaneous to the training data. In other words, the neural network may be trained to encode a relationship between an environment relative to an autonomous vehicle **100** as indicated in sensor **212** data and the corresponding control operations effected by a user or operation of the autonomous vehicle. The training module **614** may provide a corpus of training data, or a selected subset of training data, to train the neural network. For example, the training module **614** may select particular subsets of training data associated with particular driving conditions, environment states, etc. to train the neural network.

(46) The software resources **613** may include, for example, one or more modules of computer program instructions that when executed by processing resources **612** within the execution environment **227** are useful in deploying software resources or other data to autonomous vehicles **100** via a network **618**. For example, a deployment module **616** may provide software updates, neural network updates, or other data to autonomous vehicles **100** to facilitate autonomous vehicle control operations.

(47) The software resources **613** may include, for example, one or more modules of computer program instructions that when executed by processing resources **612** within the execution environment **227** are useful in collecting data from autonomous vehicles **100** via a network **618**. For example, a data collection module **620** may receive, from autonomous vehicles **100**, collected sensor **212**, associated control operations, software performance logs, or other data. Such data may facilitate training of neural networks via the training module **614** or stored using storage resources **608**.

(48) For further explanation, FIG. **7** sets forth a flow chart illustrating an exemplary method for throughput reduction in high resolution camera sensors according to some embodiments of the present disclosure. The method of FIG. **7** may be implemented, for example, in a camera sensor **700** of an autonomous vehicle **100**. The camera sensor **700** may include a sensor **212** of an autonomous vehicle **100** similar to those as set forth above. In some embodiments, the camera sensor **700** is included in a plurality of camera sensors **700** of the autonomous vehicle **100**. The method of FIG. **7** may then be performed by each camera sensor **700** of the autonomous vehicle **100**. For example, in some embodiments, the method of FIG. **7** may be performed by a pair of camera sensors **700** for a particular facing of the autonomous vehicle **100** (e.g., a pair of front facing, rear facing, or side facing camera sensors **700**). In some embodiments, the plurality of camera sensors **700** includes a plurality of camera sensor **700** pairs, with each pair corresponding to a particular facing of the autonomous vehicle **100**. Although the following discussion describes throughput reduction in the context of an autonomous vehicle camera sensor **700**, it is understood that the approaches described herein are applicable to other high resolution camera sensors not necessarily deployed in an autonomous vehicle.

(49) The method of FIG. **7** includes generating **702**, by a camera sensor **700** of an autonomous vehicle **100**, a frame **704**. The frame **704** is a digitally encoded image captured by the camera sensor **700**. For example, the frame **704** may be generated **702** during operation of the autonomous vehicle **100** and may be included in multiple frames **704** of video data.

(50) The method of FIG. **7** also includes selecting **706** an area of focus for the frame **706**. The area of focus for the frame is a subset of pixels of the frame **706** to be included in a cropped frame **712** as described below. The selected area is an area of focus in the sense that it is determined to have

particular objects, attributes, and the like that are of particular importance when generating driving decisions by an automation computing system **116**, or when performing other computational actions by the automation computing system **116** based on captured image or video data. In other words, the area of focus may correspond to a particular area of interest for the frame **704**. As the camera sensor **700** need not have telescopic, adjustable, or moveable focal lenses, the area of focus does not correspond to a variable optical focal point for the camera sensor. For example, assuming a stationary camera sensor **700** with a fixed, centralized focal point, the area of focus can include any computationally subset of a frame **704** captured by the camera sensor **700**.

(51) In some embodiments, the area of focus may be selected **706** based on one or more models. The one or models may include machine learning models, algorithmic models, rule-based models, and the like, or combinations thereof. The one or more models may accept, as input, the frame **704**, one or more previously generated frames **704**, one or more previously generated downsampled frames **710** or cropped frames **712** (described in further detail below), or combinations thereof. For example, in some embodiments, the one or more models may include a motion identifying model that identifies, based on its input, a particular area of motion within a frame **704**. The particular area of motion may include areas of motion having a degree of motion exceeding a threshold, an area of motion having a highest degree of motion, and the like. The one or more models may also include models that identify areas of highest spatial density, object density, and the like. One skilled in the art will appreciate that a variety of models or other approaches may be used to select **706** an area of focus, and that particular approaches may vary or be configured based on particular design considerations.

(52) In some embodiments, the one or more models may be executed within the camera sensor **700**. For example, after generating **702** the frame **704**, the camera sensor **700** provides the frame **704** to the one or more models to identify and select **706** the area of focus. In other embodiments, the one or more models may be executed within an automation computing system **116** of the autonomous vehicle **100**. For example, as will be described in further detail below, the camera sensor **700** may provide a downsampled frame **710** and a cropped frame **712** to the automation computing system **116**. The automation computing system **116** may provide the downsampled frame **710** and potentially other previously generated downsampled frames **710** to the one or more models to select **706** an area of focus. The automation computing system **116** may then provide, back to the camera sensor **700**, an indication of the area of focus. The camera sensor **700** then selects **706** the area of focus for a next generated frame **704** to be the area of focus identified based on the previously provided downsampled frame(s) **710**.

(53) The method of FIG. 7 also includes generating **708** (e.g., based on the frame **704**) a downsampled frame **710** and a cropped frame **712**. The downsampled frame **710** is a lower resolution version of the frame **704**. In other words, the downsampled frame **710** is generated **708** by applying a down sampling algorithm as would be understood by one skilled in the art to the frame **704**. The cropped frame **712** is a sub-area of the frame **704** (e.g., a subset of pixels of the frame **704**) corresponding to or including the selected **706** area of focus. In some embodiments, the area of focus may be selected **706** as having a fixed size or resolution. In some embodiments, the area of focus may be of variable resolution. Furthermore, in some embodiments, the cropped frame **712** may be of fixed or variable resolution (e.g., across cropped frames **712** generated for different frames **704**).

(54) For example, in some embodiments, the cropped frame **712** may be generated **708** by selecting, as the cropped frame **712**, those pixels included in the area of focus. Thus, the cropped frame **712** matches or corresponds to the area of focus. In some embodiments, the cropped frame **712** may be generated by selecting more or less pixels than those included in the area of focus. For example, where the cropped frame **712** is of a fixed size and the area of focus is of lesser size, the cropped frame **712** may be selected and generated **708** by adding in additional rows or columns of pixels to meet the resolution of the cropped frame **712**. As a further example, where the area of

focus is selected **706** as a non-rectangular region of pixels, the cropped frame **712** may be generated **708** by selecting a bounding box that encompasses or partially encompasses the non-rectangular region of pixels for the area of focus, with the cropped frame **712** including the pixels of the frame **704** encompassed by the bounding box.

(55) The method of FIG. 7 also includes providing **714** the downsampled frame **710** and the cropped frame **712** to an automation computing system **116** of the autonomous vehicle **100** instead of the frame **704**. The automation computing system **116** may then use the cropped frame **712** and downsampled frame **710** to perform various driving decisions as described above. For example, in some embodiments, the cropped frame **712** and downsampled frame **710** are provided as inputs to one or more models for generating driving decisions by the automation computing system **116**.

(56) In other embodiments, the automation computing system **116** may generate a new frame based on the downsampled frame **710** and the cropped frame **712**. For example, the automation computing system **116** may upscale the downsampled frame **710** to a higher resolution, such as the resolution at which the frame **704** was originally captured. The resulting upscaled frame would be of higher resolution than the downsampled frame **710** but lacking the fidelity of the originally generated frame **704** due to the previously applied downsampling process.

(57) Accordingly, the automation computing system **116** then maps the cropped frame **712** onto the upscaled downsampled frame **710** such that the new frame includes the pixels of the cropped frame **712** at their respective positions of the originally generated frame **704**. Thus, the resulting new frame includes an area of higher fidelity based on the cropped frame **712**, with remaining portions of the new frame being of lower fidelity. The new frame may then be provided as an input to one or more models for generating driving decisions or performing other functions of the automation computing system **116**.

(58) Where the camera sensor **700** captures the frame **704** at a high resolution, the camera sensor **700** is required to send a larger amount of data to the automation computing system **116** compared to frames **704** at a lower resolution. The data links between camera sensors **700** and the automation computing system **116** would undergo a significant burden if each camera sensor **700** provides high resolution frames **704**, particularly as additional camera sensors **700** are used throughout the autonomous vehicle **100**. Though lower resolution camera sensors **700** may be used to reduce the overall data throughput from camera sensors **700**, these lower resolution camera sensors **700** have a reduced perceptible distance compared to higher resolution camera sensors **700**. For example, a twelve megapixel camera may have a perceptible distance of around fifty meters. A forty-eight megapixel camera may have an increased perceptible distance of around seventy-five meters, but the resulting data is quadrupled in size relative to the twelve megapixel camera.

(59) The approaches set forth reduce the overall amount of data transferred between camera sensors **700** and the automation computing system **116** as the total size of the downsampled frame **710** and cropped frame may be less than the size of the original frame **704**. As the area of interest is encoded in the higher fidelity cropped frame **712**, the automation computing system **116** still makes driving decisions using image data having high fidelity at particular regions of interest. Thus, reduced throughput is achieved without negatively impacting the fidelity of the area of focus or region of interest.

(60) One skilled in the art will appreciate that the approaches set forth above with respect to FIG. 7 may be repeatedly performed during operation of the autonomous vehicle **100**. Moreover, one skilled in the art will appreciate that the method of FIG. 7 may be repeatedly performed by each of multiple camera sensors **700** installed in the autonomous vehicle **100**.

(61) For example, assume that a pair of camera sensors **700** are directed toward a particular facing of the autonomous vehicle **100** to provide stereoscopic vision for that particular facing. In some embodiments, each camera sensor **700** independently selects **706** areas of focus for their respective captured frames. In other embodiments, the pair of camera sensors **700** are configured to match or coordinate their areas of focus such that the corresponding cropped frames **712** capture a same or

matching area. For example, one or more models may be used to identify matching or corresponding portions of a frame or area of focus between two camera sensors **700** in order to account for variations caused by two camera sensors **700** with different placements being directed toward the same facing.

(62) One skilled in the art will appreciate that the approaches described above provide advantages over existing solutions relying on the image fidelity of optically focused cameras. For example, movable cameras with a central focal point may allow for different regions to be focused on by moving the camera. However, this increases the number of moving parts and increase the possibility of component failure. As another example, fixed cameras with a fixed central focal point are at disadvantage while a car is driving through a turn, where the regions of interest may be to one side or the other of the central focus. By using computationally defined areas of interest, the approaches set forth herein allow for variable areas of interest to be captured by the camera sensors **700**.

(63) For further explanation, FIG. **8** sets forth a flowchart of another example method for throughput reduction in high resolution camera sensors according to some embodiments of the present disclosure. The method of FIG. **8** is similar to FIG. **7** in that the method of FIG. **8** includes generating **702** (e.g., by a camera sensor **700** of an autonomous vehicle **100**) a frame **704**; selecting **706** an area of focus for the frame **704**; generating **708** a downsampled frame **710** and a cropped frame **712** (e.g., from the frame **704**); and providing **714** the downsampled frame **710** and the cropped frame **712** to the automation computing system **116**.

(64) The method of FIG. **8** differs from FIG. **7** in that the method of FIG. **8** includes selecting **802** another area of focus for the frame **704**. The other area of focus may be selected **802** using approaches similar to those set forth above with respect to selecting **706** the area of focus as described in FIG. **7**. For example, the other area of focus may be selected **802** using one or more models. The one or more models may include the same models used to select **706** the area of focus, or different models. For example, the other area of focus may be selected **802** using one or more motion models to identify another area of motion, an area of second highest motion, and the like.

(65) The method of FIG. **8** further differs from FIG. **7** in that generating **708** the downsampled frame **710** and the cropped frame **712** includes generating **804** another cropped frame based on the other area of focus. The other cropped frame may be generated **804** using similar approaches as those used to generate **708** the cropped frame **712**, but instead using the other area of focus. The cropped frame **712** and other cropped frame are shown in FIG. **8** in aggregate as cropped frames **806**.

(66) The method of FIG. **8** also includes providing **808** the other cropped frame to the automation computing system **116**. The other cropped frame may then be provided as an input to one or more models for determining driving decisions by the automation computing system **116**. In some embodiments, the other cropped frame is mapped onto the downsampled frame **710** (e.g., after upscaling the downsampled frame **710**) along with the cropped frame **712**, thereby resulting in a new frame with two high fidelity regions corresponding to each of the cropped frames **806**.

(67) One skilled in the art will appreciate that this allows for the automation computing system **116** to receive image data with multiple, high-fidelity regions of interest. This provides advantages over solutions requiring a high-fidelity region of interest to be targeted by an area of optical focus of a camera, as each area of focus would require an individual camera. Thus, for stereoscopic image data with two regions of interest, four total cameras would be required instead of the two camera sensors **700** that would be required by the approaches set forth herein.

(68) Although the above discussion describes using two areas of focus, thereby resulting in two cropped frames **806** of high fidelity, one skilled in the art will appreciate that any number of areas of interest may be determined and their corresponding cropped frames being generated.

(69) In view of the explanations set forth above, readers will recognize that the benefits of throughput reduction in high resolution camera sensors according to embodiments of the present

invention include: Improved performance of an autonomous vehicle by reducing the throughput between camera sensors and automation computing systems. Improved performance of an autonomous vehicle by allowing for variable, high fidelity regions of interest without the use of moving cameras. Improved performance of an autonomous vehicle by allowing for multiple areas of focus to be captured by a single camera.

(70) Exemplary embodiments of the present invention are described largely in the context of a fully functional computer system for throughput reduction in high resolution camera sensors. Readers of skill in the art will recognize, however, that the present invention also may be embodied in a computer program product disposed upon computer readable storage media for use with any suitable data processing system. Such computer readable storage media may be any storage medium for machine-readable information, including magnetic media, optical media, or other suitable media. Examples of such media include magnetic disks in hard drives or diskettes, compact disks for optical drives, magnetic tape, and others as will occur to those of skill in the art. Persons skilled in the art will immediately recognize that any computer system having suitable programming means will be capable of executing the steps of the method of the invention as embodied in a computer program product. Persons skilled in the art will recognize also that, although some of the exemplary embodiments described in this specification are oriented to software installed and executing on computer hardware, nevertheless, alternative embodiments implemented as firmware or as hardware are well within the scope of the present invention.

(71) The present invention may be a system, a method, and/or a computer program product. The computer program product may include a computer readable storage medium (or media) having computer readable program instructions thereon for causing a processor to carry out aspects of the present invention.

(72) The computer readable storage medium can be a tangible device that can retain and store instructions for use by an instruction execution device. The computer readable storage medium may be, for example, but is not limited to, an electronic storage device, a magnetic storage device, an optical storage device, an electromagnetic storage device, a semiconductor storage device, or any suitable combination of the foregoing. A non-exhaustive list of more specific examples of the computer readable storage medium includes the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a static random access memory (SRAM), a portable compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), a memory stick, a floppy disk, a mechanically encoded device such as punch-cards or raised structures in a groove having instructions recorded thereon, and any suitable combination of the foregoing. A computer readable storage medium, as used herein, is not to be construed as being transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire.

(73) Computer readable program instructions described herein can be downloaded to respective computing/processing devices from a computer readable storage medium or to an external computer or external storage device via a network, for example, the Internet, a local area network, a wide area network and/or a wireless network. The network may comprise copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and/or edge servers. A network adapter card or network interface in each computing/processing device receives computer readable program instructions from the network and forwards the computer readable program instructions for storage in a computer readable storage medium within the respective computing/processing device.

(74) Computer readable program instructions for carrying out operations of the present invention may be assembler instructions, instruction-set-architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, firmware instructions, state-setting data, or either

source code or object code written in any combination of one or more programming languages, including an object oriented programming language such as Smalltalk, C++ or the like, and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The computer readable program instructions may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider). In some embodiments, electronic circuitry including, for example, programmable logic circuitry, field-programmable gate arrays (FPGA), or programmable logic arrays (PLA) may execute the computer readable program instructions by utilizing state information of the computer readable program instructions to personalize the electronic circuitry, in order to perform aspects of the present invention.

(75) Aspects of the present invention are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), and computer program products according to embodiments of the invention. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer readable program instructions.

(76) These computer readable program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. These computer readable program instructions may also be stored in a computer readable storage medium that can direct a computer, a programmable data processing apparatus, and/or other devices to function in a particular manner, such that the computer readable storage medium having instructions stored therein comprises an article of manufacture including instructions which implement aspects of the function/act specified in the flowchart and/or block diagram block or blocks.

(77) The computer readable program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process, such that the instructions which execute on the computer, other programmable apparatus, or other device implement the functions/acts specified in the flowchart and/or block diagram block or blocks.

(78) The flowchart and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments of the present invention. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of instructions, which comprises one or more executable instructions for implementing the specified logical function(s).

(79) In some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts or carry out combinations of special purpose hardware and computer instructions.

(80) It will be understood that any of the functionality or approaches set forth herein may be facilitated at least in part by artificial intelligence applications, including machine learning

applications, big data analytics applications, deep learning, and other techniques. Applications of such techniques may include: machine and vehicular object detection, identification and avoidance; visual recognition, classification and tagging; algorithmic financial trading strategy performance management; simultaneous localization and mapping; predictive maintenance of high-value machinery; prevention against cyber security threats, expertise automation; image recognition and classification; question answering; robotics; text analytics (extraction, classification) and text generation and translation; and many others.

(81) It will be understood from the foregoing description that modifications and changes may be made in various embodiments of the present invention without departing from its true spirit. The descriptions in this specification are for purposes of illustration only and are not to be construed in a limiting sense. The scope of the present invention is limited only by the language of the following claims.

Claims

1. A method, comprising: selecting, by a first camera of an autonomous vehicle, a first area of focus for a first frame generated by the first camera; selecting, by a second camera of the autonomous vehicle, a second area of focus for a second frame generated by the second camera by matching the first area of focus for the first frame; generating, by the first camera from the first frame, a first downsampled frame and a first cropped frame, wherein the first cropped frame is based on the first area of focus; and generating, by the second camera from the second frame, a second downsampled frame and a second cropped frame, wherein the second cropped frame is based on the second area of focus.
2. The method of claim 1, wherein selecting the second area of focus comprises selecting, based on one or more models, as the second area of focus, a portion of the second frame matching the first area of focus.
3. The method of claim 1, wherein the first camera and the second camera are in a stereoscopic configuration.
4. The method of claim 1, further comprising: providing, to an automation computing system of the autonomous vehicle, the first downsampled frame and the first cropped frame instead of the first frame; and providing, to the automation computing system of the autonomous vehicle, the second downsampled frame and the second cropped frame instead of the second frame.
5. The method of claim 4, wherein the first downsampled frame and the first cropped frame are provided via a first data link having the first camera at a first end and the automation computing system at a second end, and wherein the second downsampled frame and the second cropped frame are provided via a second data link having the second camera at a third end and the automation computing system at a fourth end.
6. The method of claim 1, wherein the first camera and the second camera are included in a pair of cameras of the autonomous vehicle, and wherein the autonomous vehicle comprises a plurality of pairs of cameras including the pair of cameras.
7. The method of claim 6, wherein each pair of cameras of the plurality of pairs of cameras correspond to a different facing of the autonomous vehicle.
8. The method of claim 1, wherein selecting the first area of focus comprises selecting, as the first area of focus, an area of the first frame having a highest spatial density.
9. The method of claim 1, wherein selecting the first area of focus comprises selecting, as the first area of focus, an area of the first frame having a highest object density.
10. The method of claim 1, wherein selecting the first area of focus comprises selecting, as the first area of focus, an area of the first frame having one or more of: a degree of motion exceeding a threshold or a highest degree of motion.
11. An autonomous vehicle, comprising: an automation computing system; a first camera

configured to: select a first area of focus for a first frame generated by the first camera; generate a first downsampled frame and a first cropped frame, wherein the first cropped frame is based on the first area of focus; a second camera configured to: select a second area of focus for a second frame generated by the second camera by matching the first area of focus for the first frame; and generate a second downsampled frame and a second cropped frame, wherein the second cropped frame is based on the second area of focus.

12. The autonomous vehicle of claim 11, wherein selecting the second area of focus comprises selecting, based on one or more models, as the second area of focus, a portion of the second frame matching the first area of focus.

13. The autonomous vehicle of claim 11, wherein the first camera and the second camera are in a stereoscopic configuration.

14. The autonomous vehicle of claim 11: wherein the first camera is further configured to provide, to an automation computing system of the autonomous vehicle, the first downsampled frame and the first cropped frame instead of the first frame; and wherein the second camera is further configured to provide, to the automation computing system of the autonomous vehicle, the second downsampled frame and the second cropped frame instead of the second frame.

15. The autonomous vehicle of claim 14, wherein the first downsampled frame and the first cropped frame are provided via a first data link having the first camera at a first end and the automation computing system at a second end, and wherein the second downsampled frame and the second cropped frame are provided via a second data link having the second camera at a third end and the automation computing system at a fourth end.

16. The autonomous vehicle of claim 11, wherein the first camera and the second camera are included in a pair of cameras of the autonomous vehicle, and wherein the autonomous vehicle comprises a plurality of pairs of cameras including the pair of cameras.

17. The autonomous vehicle of claim 16, wherein each pair of cameras of the plurality of pairs of cameras correspond to a different facing of the autonomous vehicle.

18. The autonomous vehicle of claim 11, wherein selecting the first area of focus comprises selecting, as the first area of focus, an area of the first frame having a highest spatial density.

19. The autonomous vehicle of claim 11, wherein selecting the first area of focus comprises selecting, as the first area of focus, an area of the first frame having a highest object density.

20. The autonomous vehicle of claim 11, wherein selecting the first area of focus comprises selecting, as the first area of focus, an area of the first frame having one or more of: a degree of motion exceeding a threshold or a highest degree of motion.
